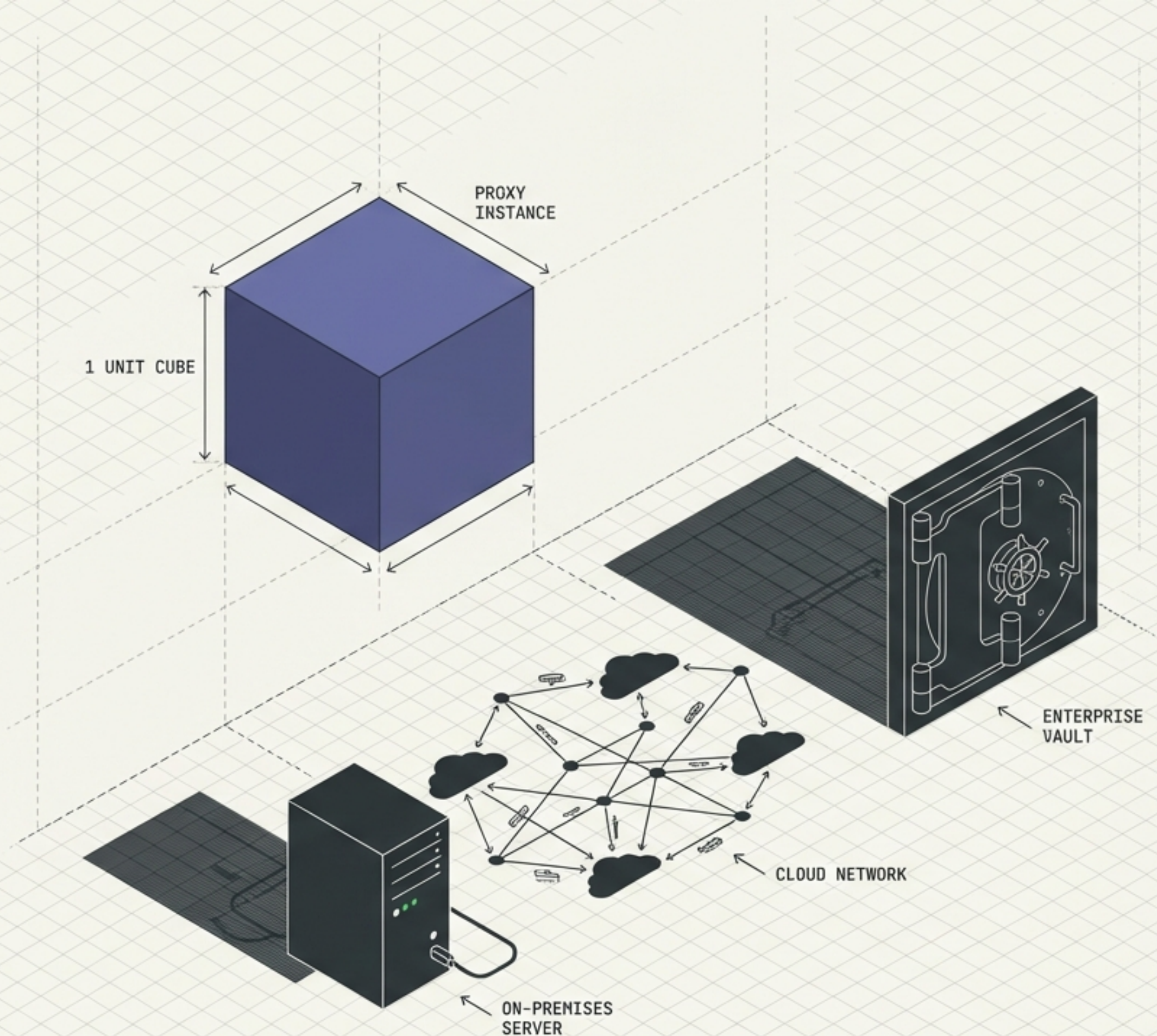




Deployment Modes for the MiTM Proxy Solution

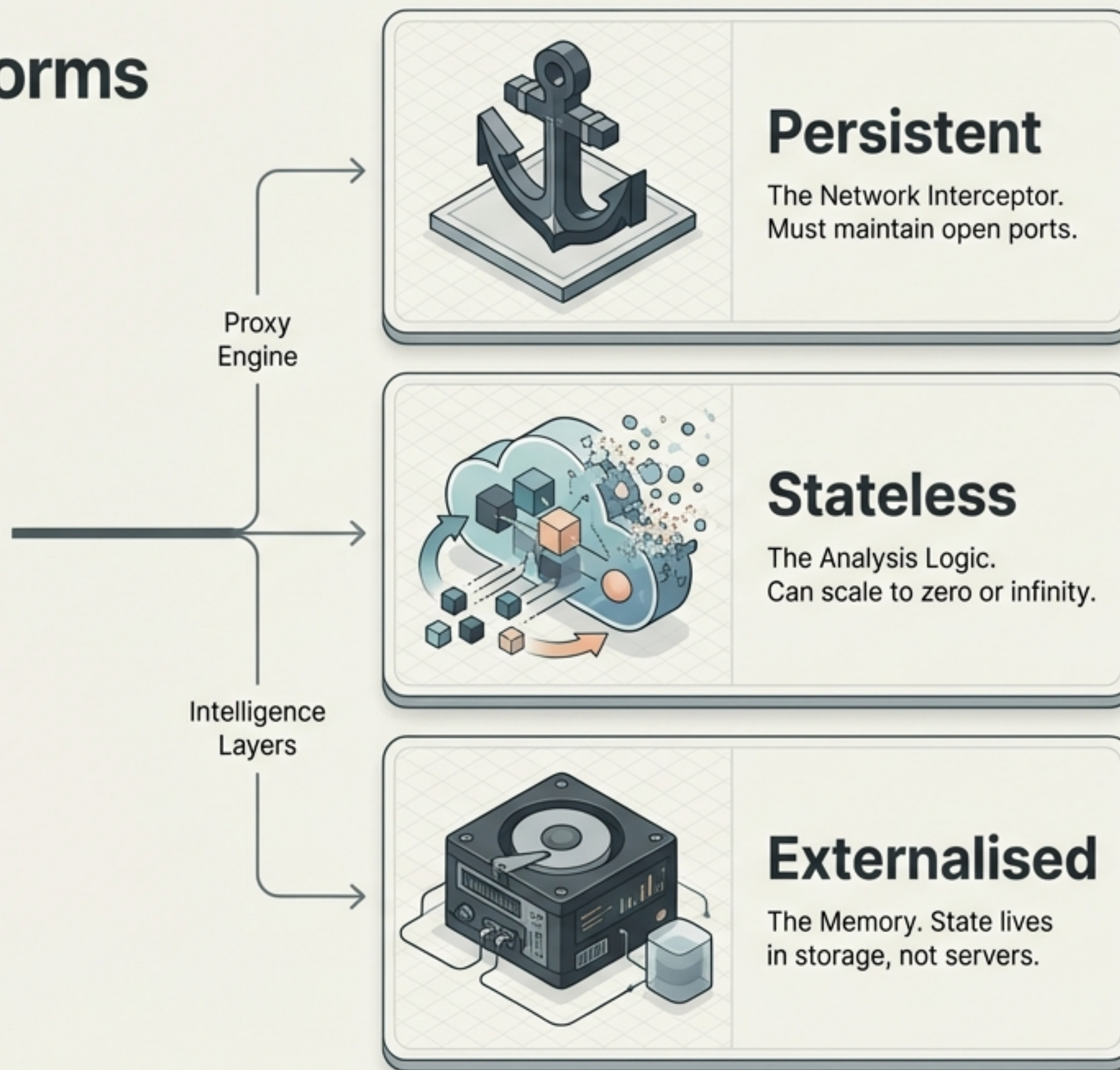
A Guide to Architecting Scalable, Cloud-Agnostic Proxy Solutions on AWS and Beyond.

An architectural guide for Developers and DevOps Engineers.

One Architecture, Multiple Forms

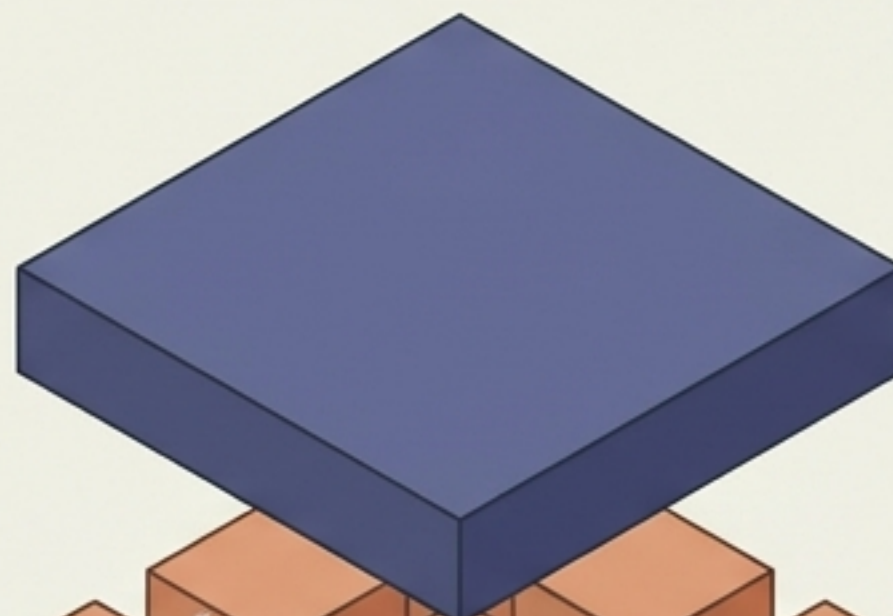
The solution is designed to adapt to the environment—whether it's a laptop, a secure air-gapped facility, or a hyper-scale cloud environment.

By decoupling the persistent Proxy Engine from the stateless Intelligence Layers, the system allows for massive flexibility without code rewrites.



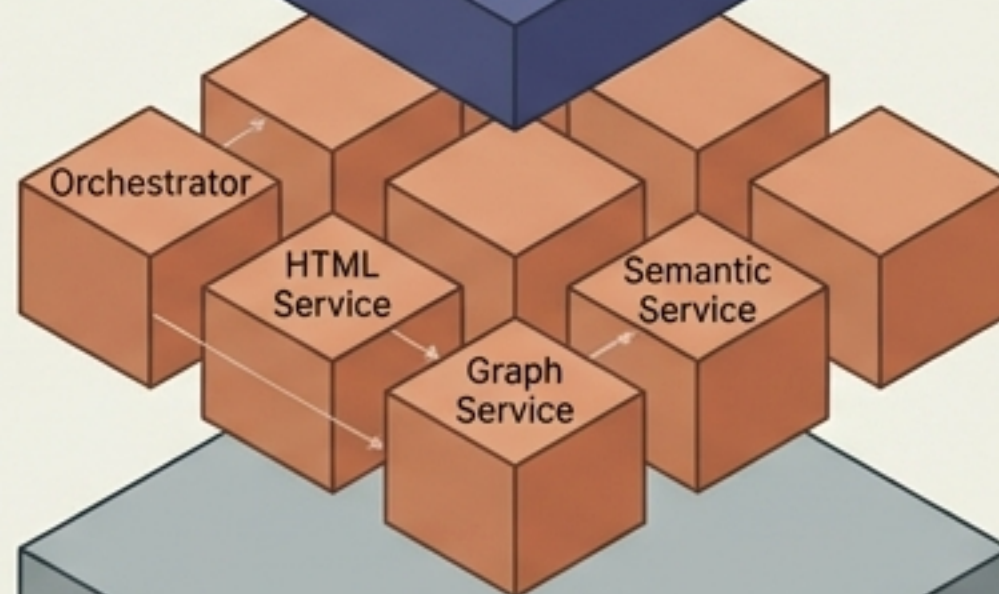
The Anatomy of the Solution

1. The Interceptor



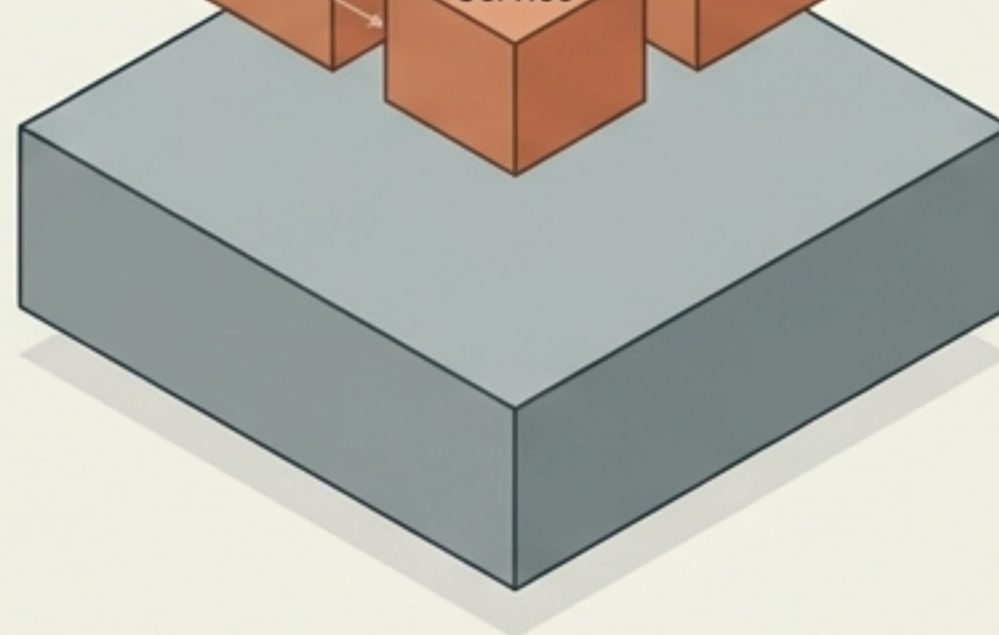
MiTM Proxy Engine.
Persistent HTTP/HTTPS proxy.
Handles TLS interception.

2. The Brains



Microservice APIs (FastAPI).

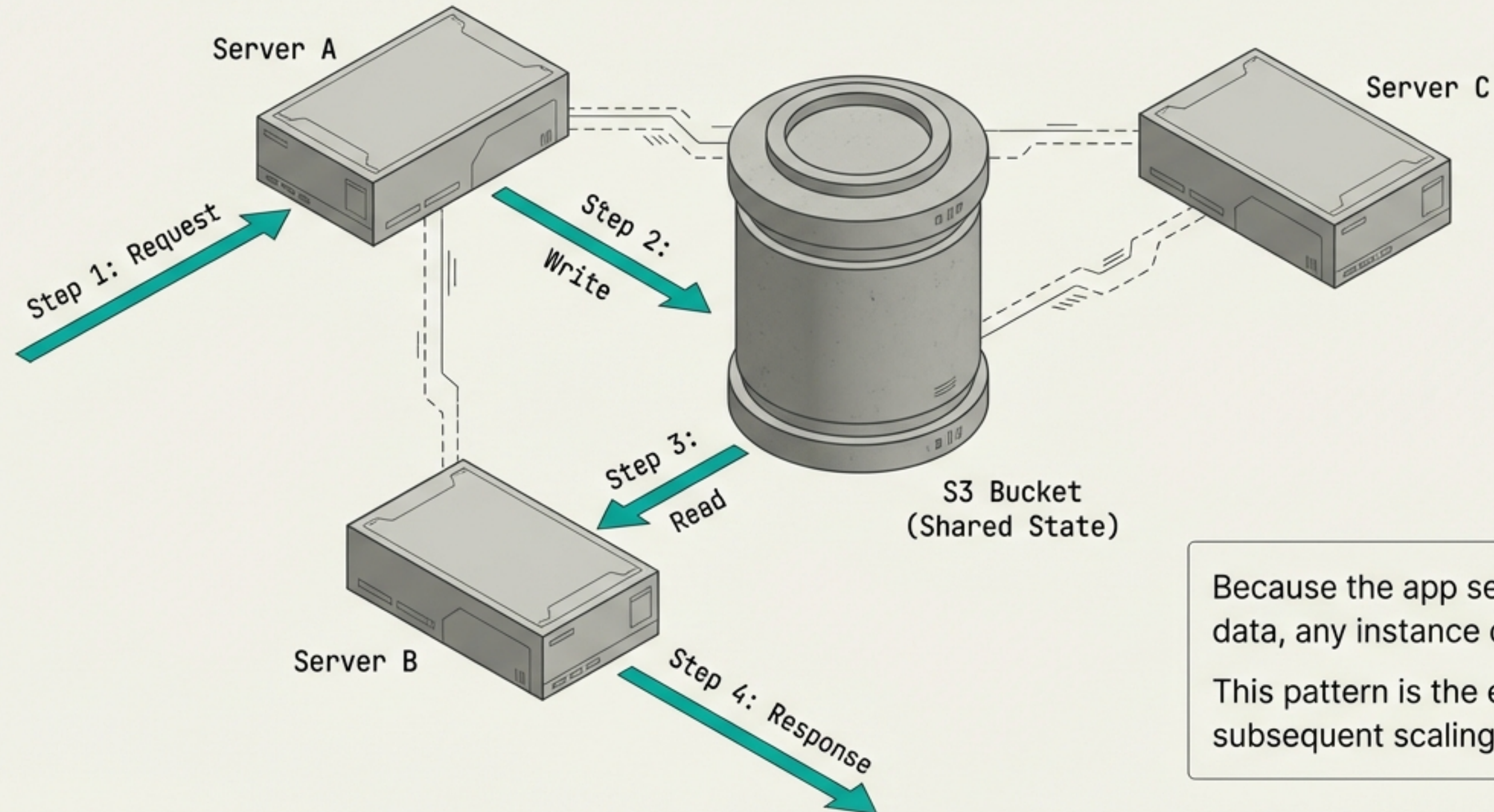
3. The Memory



Storage Backend (S3/Object Storage).
Stores pages, graphs, and analysis results.

The Golden Rule: Externalised State

How do we switch deployment modes without rewriting code?

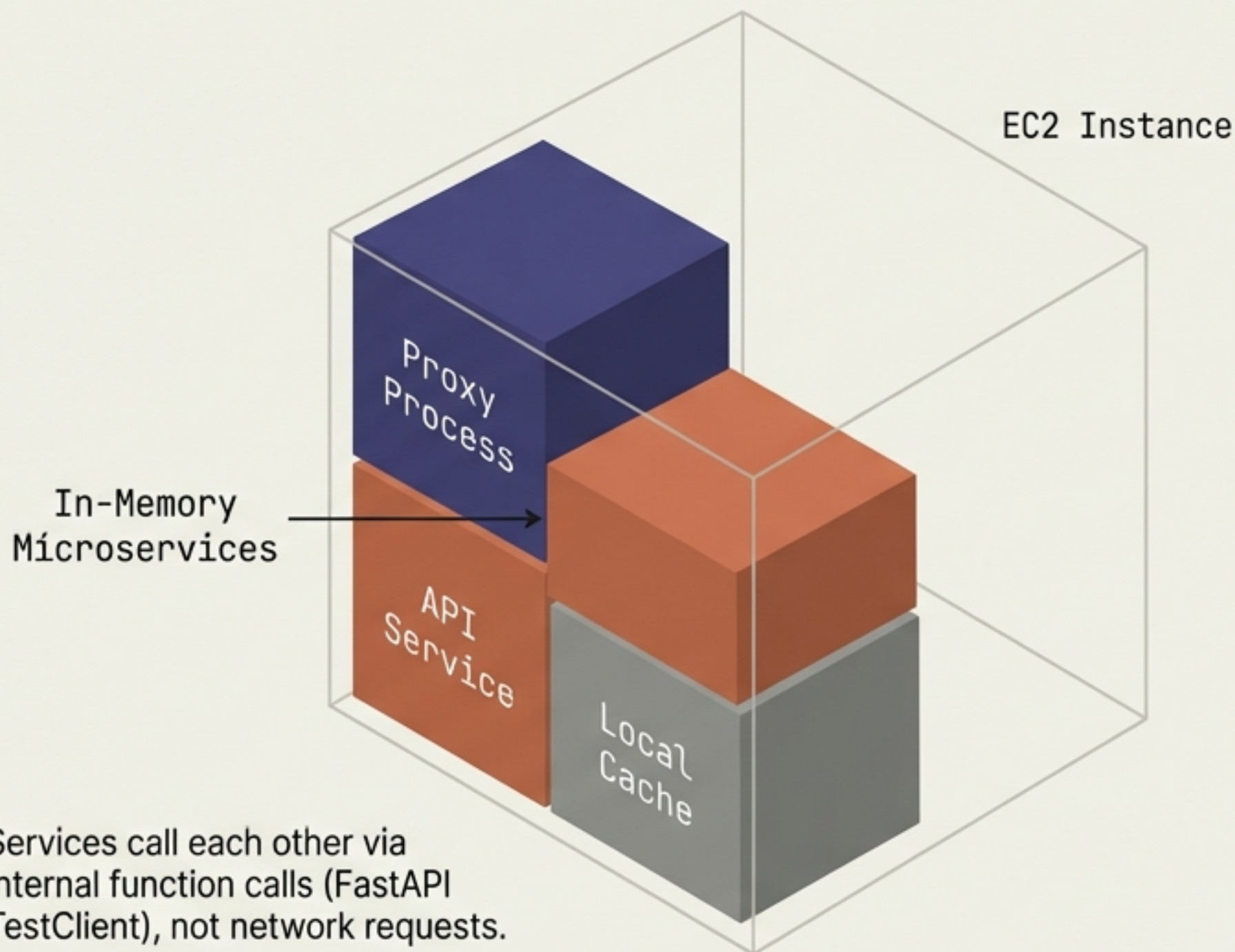


Mode 1: The All-in-One Monolith

Scenario: Development, Testing, Appliance delivery.

Architecture: Everything runs on a single EC2 instance.

Why choose it: Zero network latency, simplest operations, easiest debugging.

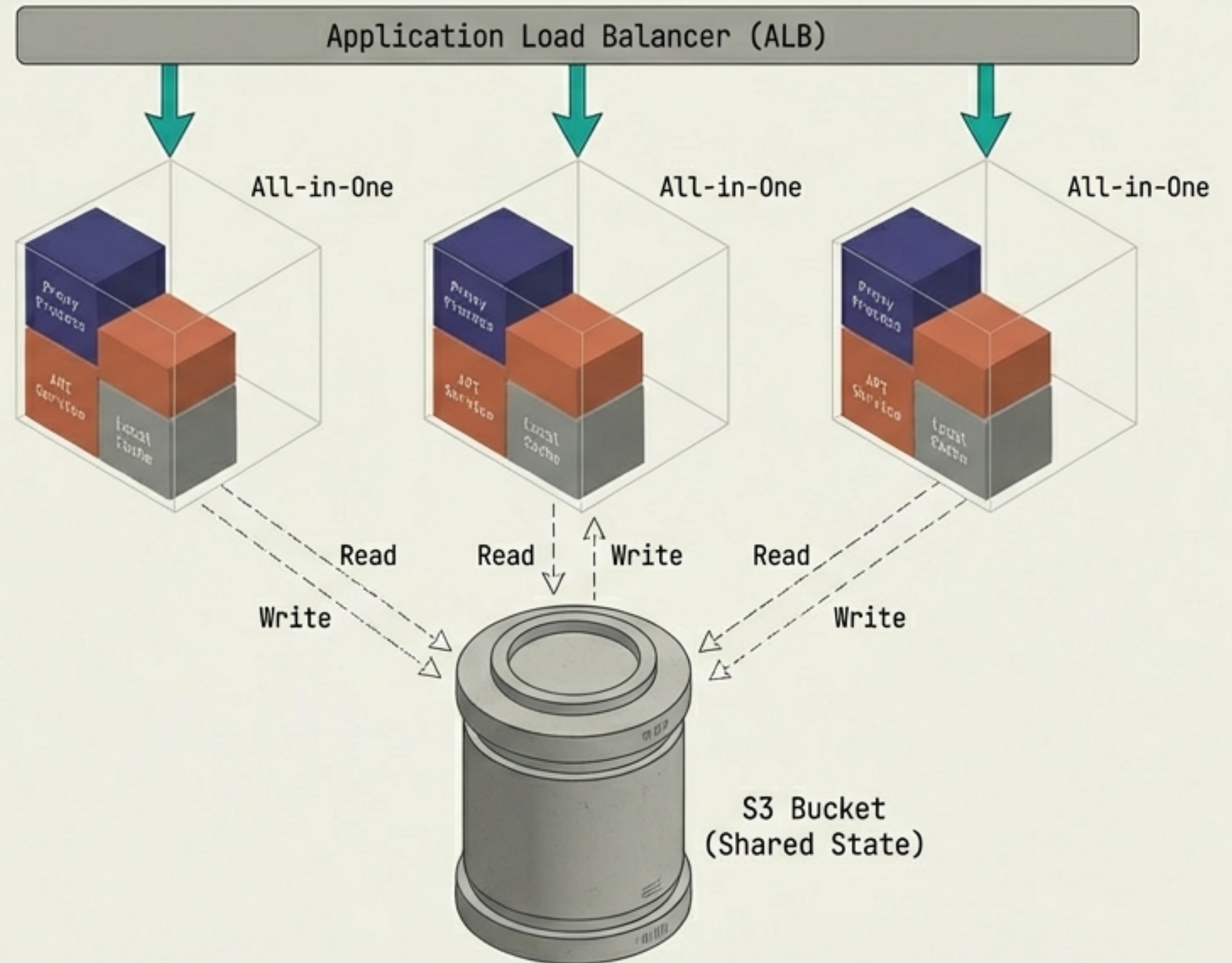


Scaling Mode 1: The Replicated Monolith

Scenario: Moderate traffic,
High Availability
requirements.

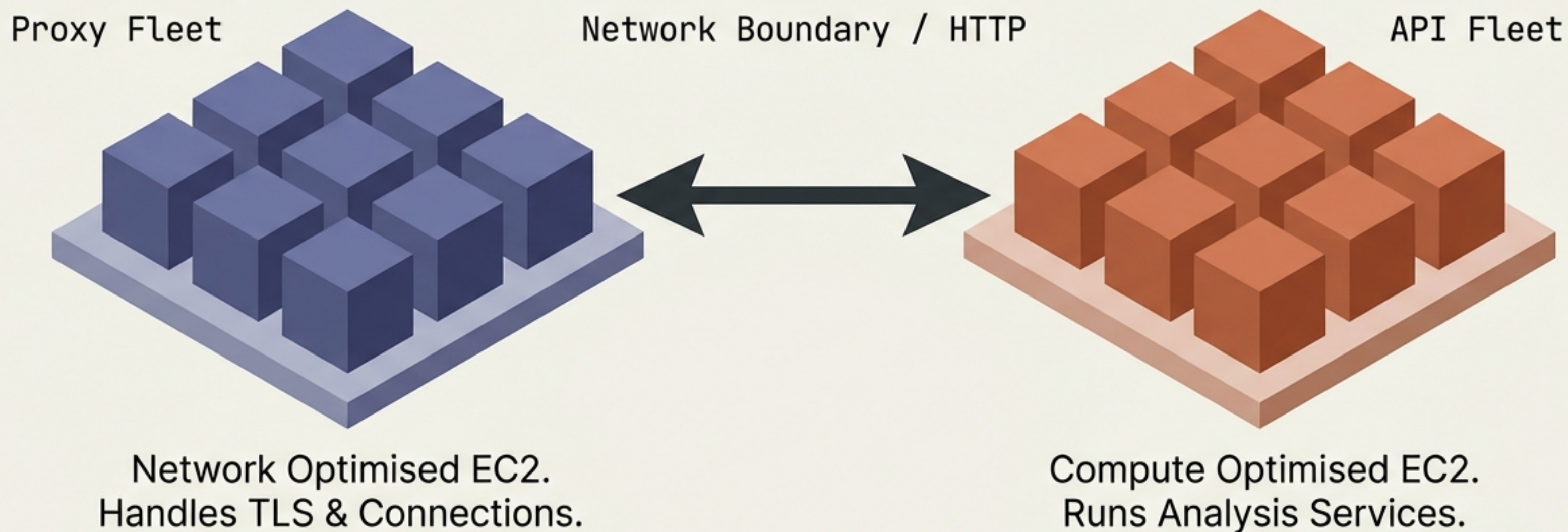
Strategy: Horizontal scaling
(more instances) vs Vertical
scaling.

Critical Dependency: Shared
storage ensures
consistency.



Mode 2: The Two-Tier Split

Decoupling Network from Compute

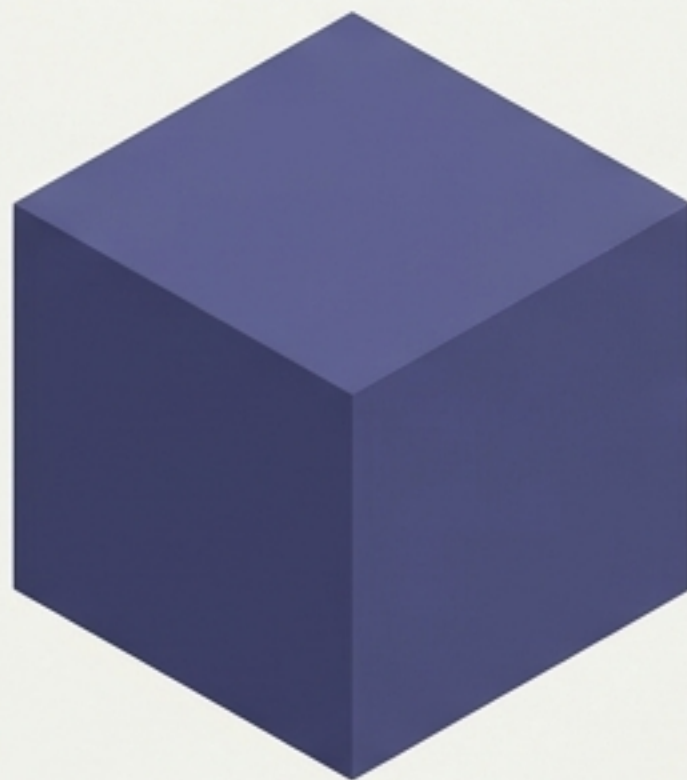


Trade-off: Increases flexibility for security segmentation (DMZ vs Private) but introduces mild network latency.

Mode 3: The Hybrid Serverless

Reducing Operational Overhead

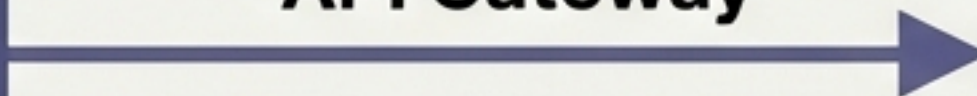
Persistent EC2 Proxy



Single Lambda Function



API Gateway

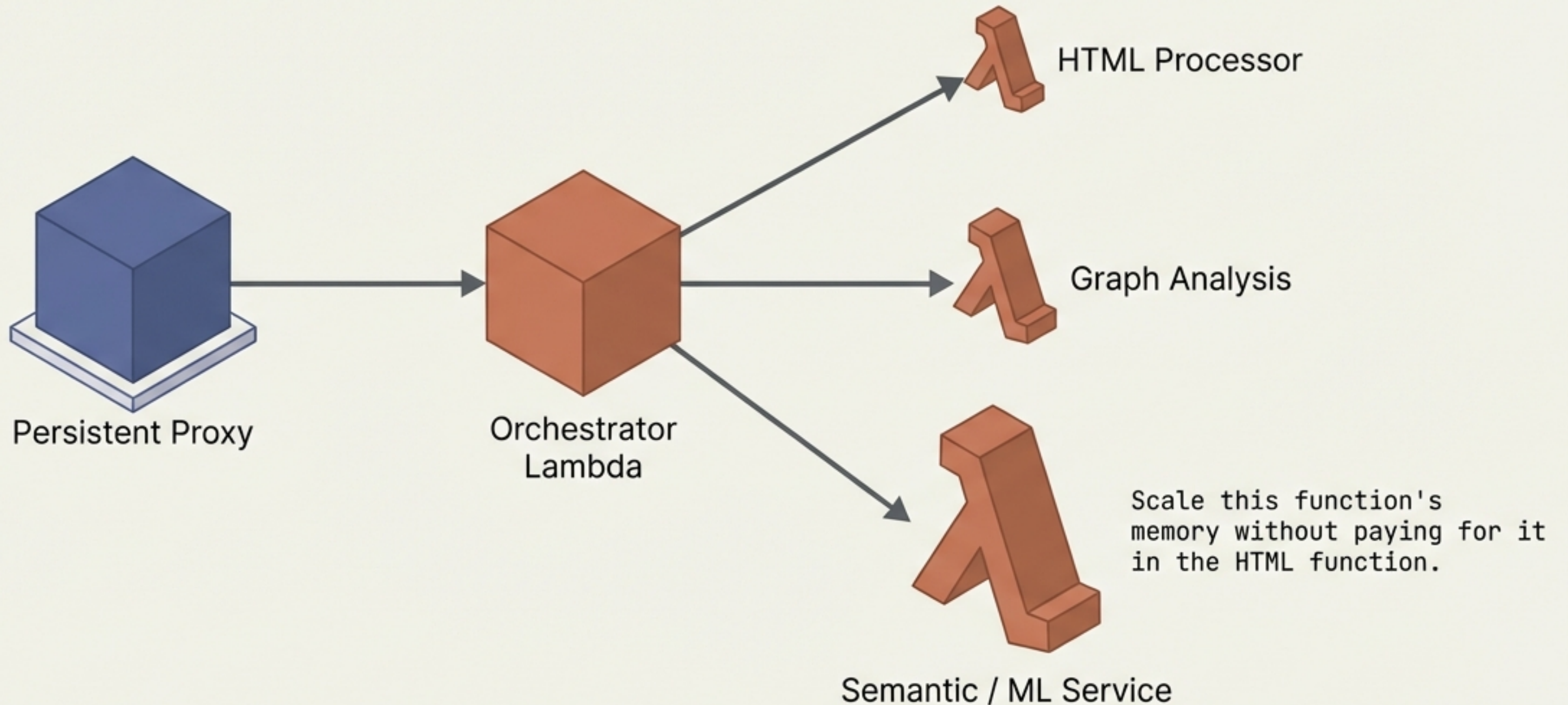


All APIs
(HTML + Graph
+ Semantic)

- The API layer scales to zero when not in use.
- Cost-efficient for bursty traffic.
- Constraints: Cold starts and 15-minute execution limits.

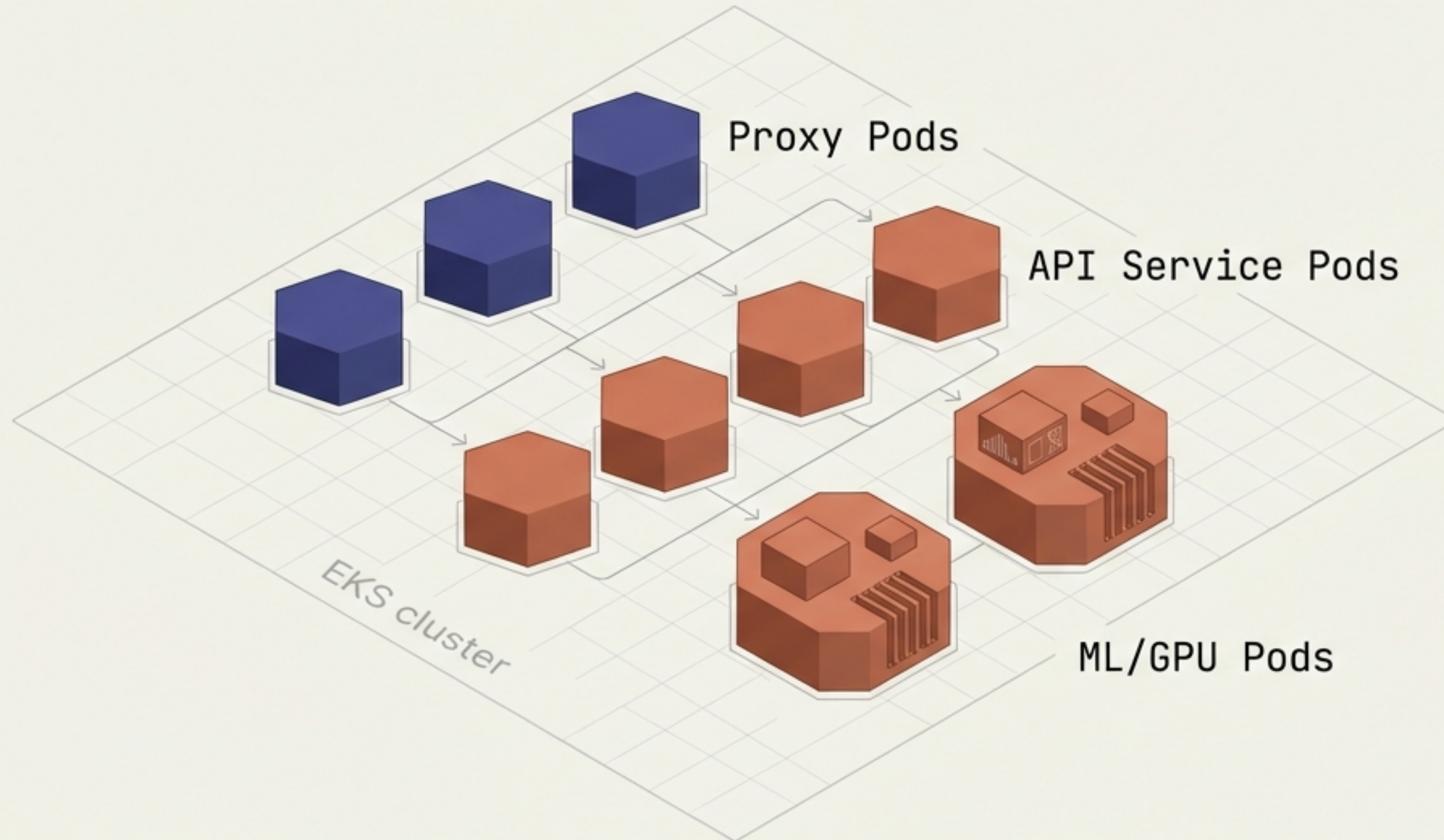
Mode 4: Fully Serverless Microservices

Maximum granularity and cloud-native purity. Isolate failures and scale functions independently.



Mode 5: The Container Orchestrator (Kubernetes)

Scale without Serverless limits.



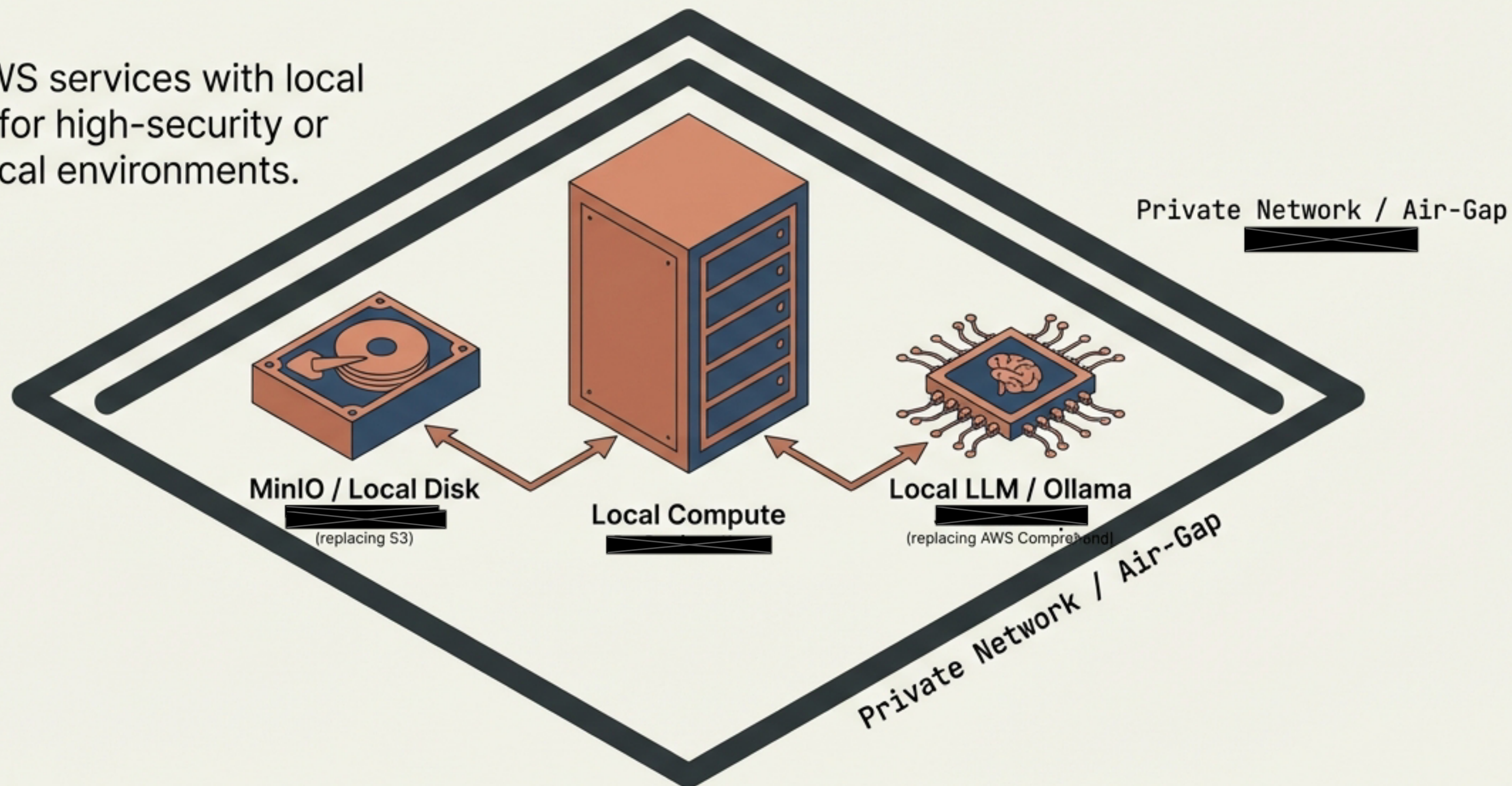
Why choose it:

- Long-running tasks (no timeout).
- Cloud portability.
- Access to specialised hardware (GPUs).

Mode 6: The Fortress (Offline / Air-Gapped)

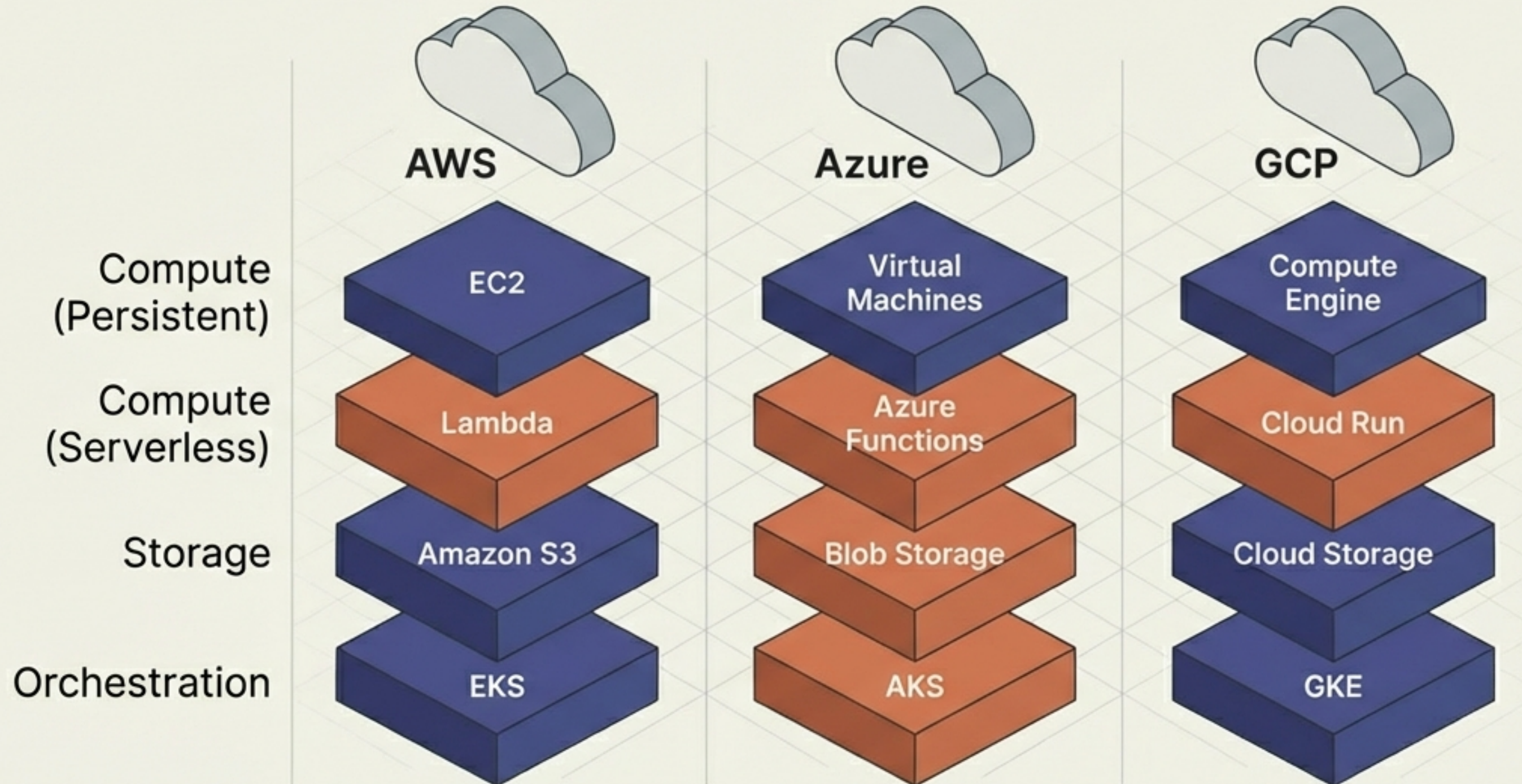
When the cloud is not an option.

Replaces AWS services with local equivalents for high-security or privacy-critical environments.



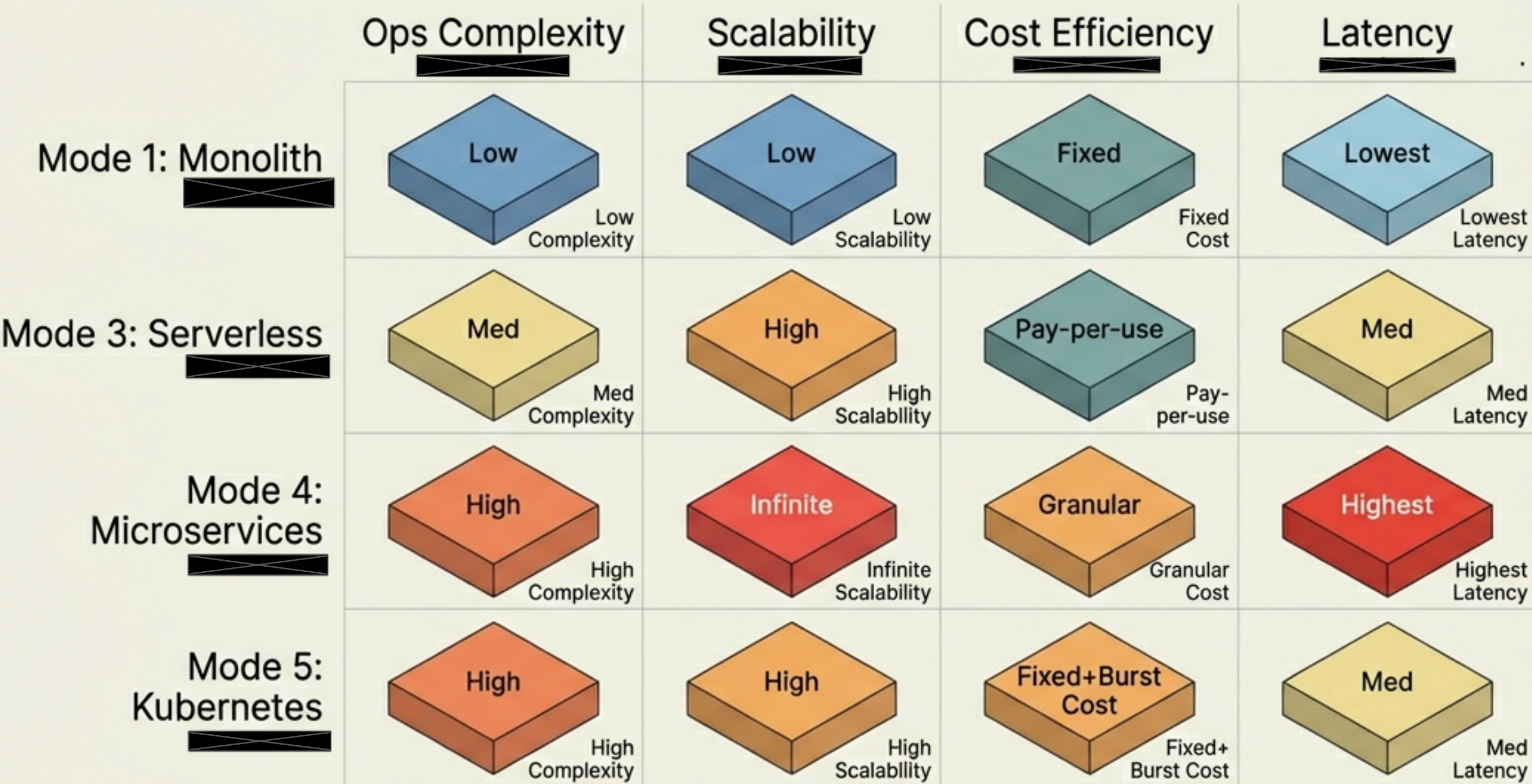
Cloud Agnosticism

The patterns are universal; only the names change.

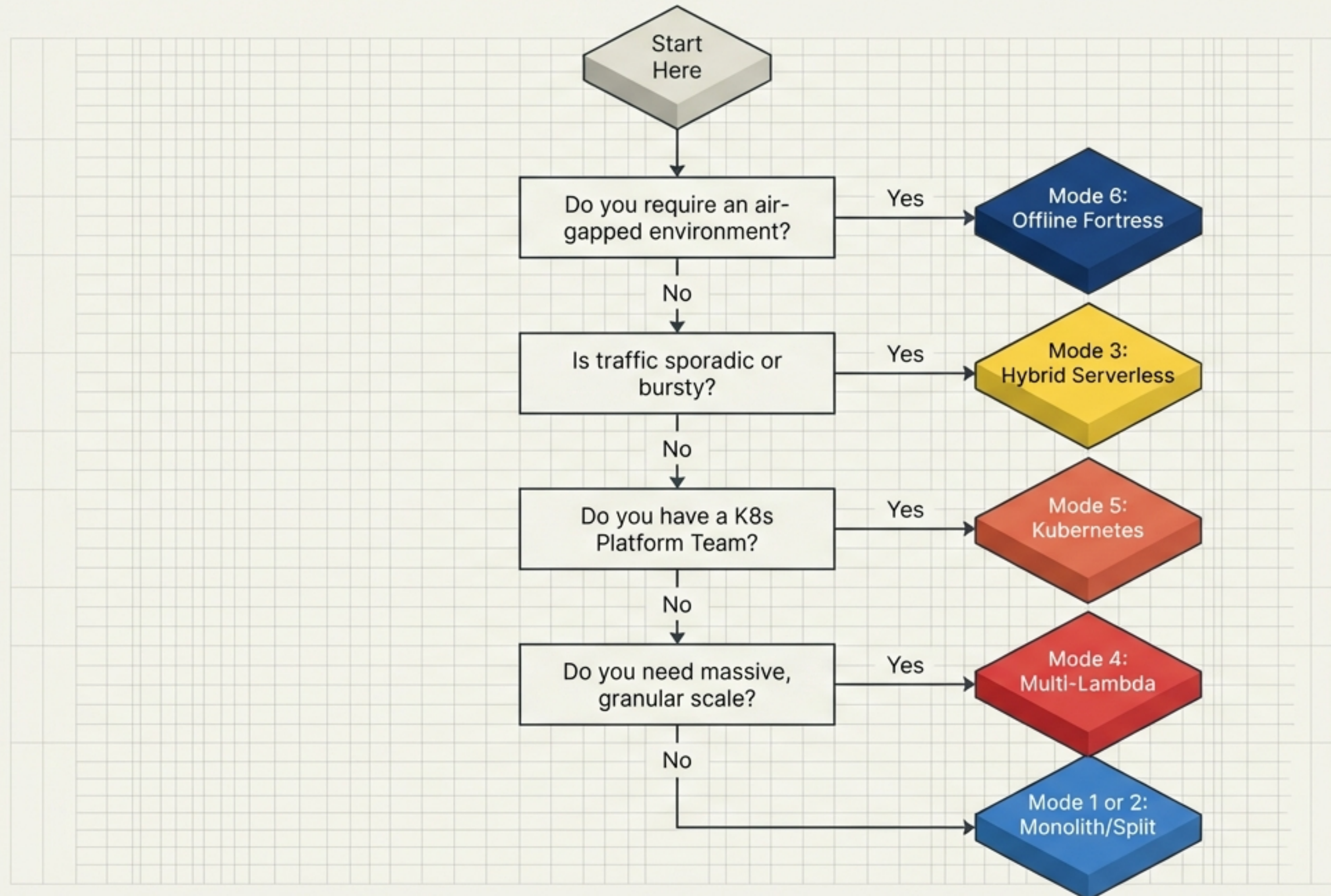


Comparative Analysis: The Trade-off Matrix

Visualizing the benefits and challenges of different architectural modes.

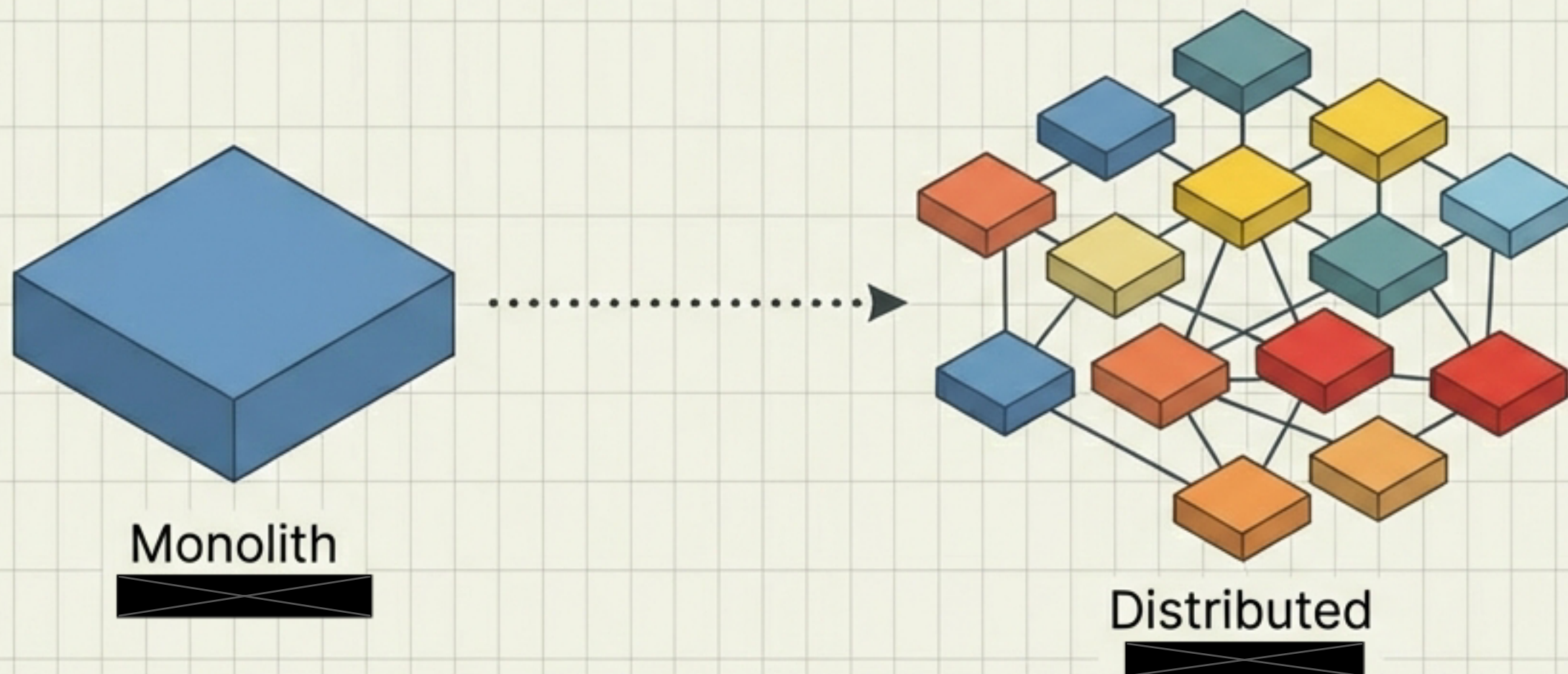


Choosing Your Architecture



Future-Proof by Design

The best architecture is the one that fits your current constraints. Start simple, knowing the path to evolution is paved by the design.



Evaluate your traffic patterns and operational capacity, then deploy the mode that matches your reality.